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**Project: Adaptive Key Resizing to Reduce Errors**

Abstract:

For many people typing on a smaller touchscreen keyboard proves challenging for them for a plethora of reasons. These reasons include but are not limited to variation in finger size, different grip postures and limited screen space, which all are factors in why users may have decreased typing speeds and recurring typos. Although predictive text and autocorrect both do a good job of reducing errors they sometimes also cause their own problems by altering words in an incorrect way. The purpose of this project is to target the layout of the keyboard itself so that the margin of error can be reduced from the source.

The proposed idea is as follows- to create an adaptive key resizing keyboard that would automatically adjust key sizes based on the user behavior of the individual. The proposed system would constantly monitor input patterns and recognize frequently mistyped characters and then enlarge those specific keys while proportionally shrinking less error prone keys. For example if the user kept on improperly typing the letter ‘x’ the interface would slightly increase the size of the letter making use of lightweight error tracking algorithms to monitor the success rate of the new slightly modified keyboard.

To evaluate the effectiveness of this proposed idea, there will be multiple controlled studies performed comparing this new adaptive keyboard to its static counterparts. The measures of efficacy will include typing speed and error rate. I hypothesize that this new adaptive keyboard resizing should reduce error rates while also maintaining typing speed.